

Agentic AI-Enabled Cross-Domain Orchestration for Space-Air-Ground Integrated Networks

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Abstract—The future sixth-generation (6G) network aims to enable intelligent space-air-ground integrated network (SAGIN) architectures that support autonomous operation and seamless global connectivity services. With the rapid development of large artificial intelligence models (LAMs), such as large language models (LLMs), agentic artificial intelligence (AI)-enabled networks have recently emerged as a promising solution that extends beyond the limited generalization capability of traditional intelligent systems for autonomous network operation and management. However, deploying multiple agents across different control domains in large-scale SAGIN systems, including the radio access network (RAN), core network (CN), heterogeneous computing infrastructures across ground-to-space layers, and end-to-end (E2E) service management, introduces challenges in aligning heterogeneous domain-level optimization objectives with global E2E service performance goals. This paper first analyzes the architectural and control-domain challenges of agentic AI-enabled SAGIN and highlights the need for intelligent cross-domain coordination. We then propose an agentic AI-enabled cross-domain orchestration framework for SAGIN (AAI-CDOS), where domain-specific agents autonomously negotiate and coordinate to align system objectives and improve the quality of service (QoS) for SAGIN applications. As a representative multi-domain optimization scenario in SAGIN for E2E applications, an urgent wildfire detection use case involving distributed model updating via SAGIN-assisted federated learning (FL) is presented to demonstrate the effectiveness of the proposed cross-domain orchestration framework. Simulation results demonstrate that AAI-CDOS enhances the performance of FL by enabling autonomous multi-agent coordination across distributed control domains, jointly optimizing radio and computing resources and achieving higher model test accuracy than baseline schemes.

Index Terms—Agentic AI, space-air-ground integrated network (SAGIN), 6G networks, cross-domain, wireless communications

I. INTRODUCTION

As space-air-ground integrated networks (SAGIN) gain increasing attention as a key architecture for sixth-generation (6G) networks, the proliferation of smart devices and network nodes across ground, aerial, and space layers creates a highly dynamic and heterogeneous environment, which introduces challenges for efficient network management and resource

coordination. While intelligent optimization approaches, such as deep reinforcement learning (DRL) and large language models (LLMs), have been applied to address these challenges, their effectiveness remains limited under highly dynamic 6G conditions. In particular, DRL methods often have limited generalization capability across rapidly changing environments, whereas LLMs, despite their flexibility, face reliability and robustness issues, such as hallucinations, which restrict their applicability to complex and dynamic SAGIN [1]–[4].

Hence, agentic artificial intelligence (AI), an emerging AI paradigm that operates proactively by autonomously acquiring network information and dynamically adapting its objectives throughout lifelong operation without continuous human intervention, provides a promising solution to these challenges [5]. Specifically, agentic AI can effectively manage the dynamic and complex nature of SAGIN by enabling autonomous perception, decision-making, and adaptive control.

Although agentic AI demonstrates strong potential for autonomously optimizing decision-making in highly dynamic SAGIN environments without human intervention, existing studies are largely limited to control mechanisms confined within a single domain, which is insufficient for managing large-scale and highly heterogeneous SAGIN systems [6]. Unlike 5G networks, which primarily focus on optimizing radio access network (RAN) and core network (CN) functions within a single control domain, 6G-enabled SAGIN requires multiple distributed control domains, including RAN control, CN management, computing resource orchestration, and end-to-end (E2E) service management across space-air-ground infrastructures. Furthermore, the increasing disaggregation and softwarization of network architectures, such as the Open RAN (O-RAN), enable independent deployment while supporting more flexible and programmable control mechanisms [7]–[9]. This architectural shift toward distributed control domains in future networks highlights the need for distributed agentic AI to enable cross-domain control, including the RAN domain for radio and mobility management, the CN domain for traffic steering and network slice orchestration, the computing infrastructure domain for resource and energy management, and the E2E service domain for intent translation and performance assurance.

While agentic AI-enabled cross-domain control allows each control domain to autonomously optimize its own objectives, the distinct characteristics and requirements of each domain, combined with independent decision-making, may lead to conflicting actions that degrade overall system performance. For example, in a computation offloading scenario for real-time video analytics of disaster monitoring or maritime surveillance

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in remote areas within a SAGIN environment, the agent in the RAN may reduce the allocated bandwidth to minimize transmission energy consumption or spectrum utilization. Meanwhile, the agent in the computing infrastructure, operating on ground edge server, unmanned aerial vehicles (UAVs) or satellites, may allocate additional computing resources to satisfy E2E latency constraints. Although each domain optimizes its own objective, these independently adapted decisions may lead to locally optimal but globally suboptimal decisions, resulting in violations of E2E service requirements. Moreover, cross-domain coordination in SAGIN is further complicated by the heterogeneous radio and computing characteristics across the ground, aerial, and space layers, where each layer experiences distinct channel conditions, computational capabilities, and latency constraints.

Therefore, although deploying multiple autonomous agents across different control domains is necessary for managing large-scale and highly heterogeneous SAGINs, a robust coordination architecture and associated mechanisms are required to ensure effective inter-domain cooperation, objective alignment, and the proper handling of complex interactions among heterogeneous networking and computing components [7].

In this paper, we present a comprehensive perspective on agentic AI-enabled network architectures for cross-domain orchestration in SAGIN, focusing on the coordinated operation of distributed agents across multiple domains, including the RAN, CN, computing infrastructure across space-air-ground layers, and E2E service management. The key contributions of this work are summarized as follows:

- We first provide an overview of the SAGIN architecture, its control domains, and intelligent control mechanisms in future networks. We then introduce the integration of agentic AI across the space, air, and ground layers to enhance E2E service performance. Finally, we present the key challenges of agentic AI-enabled SAGIN architectures and highlight the motivation for cross-domain control.
- We propose a cross-domain orchestration framework in which multiple agents operate across different network domains, including O-RANs, CNs, heterogeneous SAGIN computing nodes, and service management functions. In the proposed framework, agents in different domains, each with distinct objectives, autonomously negotiate and coordinate their decisions, align actions and goals through appropriate coordination mechanisms, and collectively enhance E2E network application performance.
- We conduct a case study on urgent model updates for wildfire detection, enabled by SAGIN-assisted federated learning (FL), to demonstrate how the proposed framework enhances E2E application performance through coordinated cross-domain multi-agent control across the ground-to-space layers. The results show that the proposed framework achieves the highest model accuracy under limited time constraints compared to the baseline approaches, thereby validating the effectiveness of the proposed cross-domain orchestration mechanism.

II. BACKGROUNDS

A. Overview of Intelligent SAGIN

1) *Heterogeneous Computing Infrastructure and Services in SAGIN*: SAGIN consists of three layers, ground, aerial, and space, each equipped with heterogeneous computing infrastructure. The ground layer includes resource-constrained devices such as vehicles and Internet of Things (IoT) nodes, as well as computing resources deployed at mobile edge computing (MEC) servers, base stations (BSs), roadside units (RSUs), and ground stations (GSs). The aerial layer comprises UAVs and high-altitude platform stations (HAPS), which support radio and computation functions. The space layer consists of satellite systems deployed at different orbital altitudes, including low Earth orbit (LEO), medium Earth orbit (MEO), and geostationary Earth orbit (GEO) satellites, with varying coverage, latency, and computing capabilities [2]. SAGIN supports a diverse set of services and applications, including resource allocation, task offloading, routing, and FL, for application scenarios such as intelligent transportation systems, disaster response, and remote sensing [6]. Each application has different performance requirements in terms of latency, reliability, energy efficiency, and computational capability, and the ground, aerial, and space layers play different roles in supporting these requirements. To manage heterogeneous computing infrastructure across different SAGIN layers for diverse applications, intelligent control mechanisms have been explored. For example, the authors in [4] model LEO satellites with onboard edge computing as intelligent agents to jointly optimize discrete task offloading decisions and continuous heterogeneous computing resource allocation, aiming to minimize system energy consumption while satisfying task delay constraints.

2) *Intelligent O-RAN in SAGIN*: To support global service continuity and reduce vendor dependence in future 6G networks, network virtualization and O-RAN architectures have been actively promoted by standardization bodies such as the third generation partnership project (3GPP). Unlike traditional tightly coupled network architectures, O-RAN adopts a disaggregated and virtualized design in which network functions are implemented as software-based virtual network functions (VNFs), enabling flexible deployment, independent function evolution, and interconnection through standardized open interfaces [8], [9]. Moreover, data-driven and intelligent O-RAN approaches have been proposed to enhance RAN control and optimize radio resource utilization and communication efficiency. The authors in [9] propose deploying multiple agents within the service management and orchestration (SMO) to manage distributed RAN components, such as central units (CUs) and distributed units (DUs), which handle time-critical PHY/MAC, transport, and user-plane functions that determine cell throughput and latency.

While intelligent O-RAN has been widely investigated, most existing approaches focus on radio-centric orchestration, assuming static backhaul connectivity and ground-centric control architectures. In contrast, future networks aim to achieve global connectivity through the integration of ground, aerial, and space layers in SAGIN. The heterogeneous and multi-

layered nature of SAGIN introduces several challenges to control and coordination. For example, satellite-to-ground gateway links frequently become bottlenecks, intra-constellation routing is highly dynamic, and control operations must tolerate intermittent or degraded connectivity to ground infrastructure. To address the limitations of ground-centered control loops, the authors in [10] propose Space-O-RAN, a distributed control architecture that extends O-RAN principles into satellite constellations through hierarchical closed-loop control. Considering the long propagation delays between satellites and ground-based RAN controllers, as well as deployment constraints across different layers, control and management functions in Space-O-RAN are not always hosted on the ground but are instead logically distributed across different layers.

3) *Limitations of Current Intelligent SAGIN*: Despite recent advances in intelligence-enabled SAGIN, most existing approaches embed intelligence within predefined control logic using learning-based models such as DRL or multi-agent DRL. Nevertheless, DRL-based approaches are typically task-specific and depend on predefined state-action spaces and reward functions, which limits their flexibility and ability to adapt to highly dynamic and partially observable SAGIN environments.

Fortunately, the rapid development of LLMs has motivated LLM-based multi-agent designs to enhance adaptability in SAGIN. For example, the authors in [3] propose an LLM-based multi-agent situation awareness framework for zero-trust SAGIN, in which multiple LLM agents collaboratively process heterogeneous threat information and generate security assessments and defense strategies under dynamic and partially observable conditions. However, such approaches are primarily designed for specific application contexts and remain tightly coupled to predefined functional pipelines and task objectives, which limits their autonomy in identifying objectives and problems and reduces their general applicability to broader intelligent SAGIN service management. Therefore, the AI paradigm is shifting toward agentic AI, which emphasizes autonomous goal identification, adaptive reasoning, and decision-making in complex SAGIN environments.

B. Overview of Agentic AI

1) *Definition of Agentic AI*: Compared with traditional AI-based communication approaches, agentic AI-enabled communication operates in a proactive manner by autonomously searching for and retrieving network information, continuously learning, and dynamically adjusting its objectives throughout lifelong operation without requiring ongoing human intervention. Specifically, the main features of agentic AI-enabled communication are summarized as follows:

- 1) *Autonomous perception and decision-making*: The primary goal of agentic AI-enabled communication is to minimize reliance on human intervention once agents are initialized. After initialization, agents continuously interact with the environment to perceive real-time network information, dynamically adjust their objectives in response to changing conditions, and autonomously

perform sequential and coordinated decision-making. Moreover, they can self-evolve as their hardware, tools, or application programming interfaces (APIs) are updated [11].

- 2) *External API integration and orchestration*: Agents interact with external APIs, software services, databases, and orchestration platforms to achieve system-level objectives. This enables agents to retrieve real-time information and trigger external services, while coordinating multi-step workflows across tools and platforms to accomplish complex objectives autonomously.
- 3) *Real-time responsiveness*: Agentic AI systems must be capable of responding promptly to real-time events, such as network traffic fluctuations, maintenance or failure of external API calls, and changes in software environments. Moreover, agents should dynamically adjust their strategies, models, or algorithms in response to such changes to maintain robust and efficient system operation.

2) *Architecture of Agentic AI*: The basic layered architecture of agentic AI consists of four layers [5], [11]. The *perception layer* enables agents to sense and interpret real-time SAGIN conditions through heterogeneous multimodal data (i.e., signals, images, texts) collection and preprocessing. The *knowledge and memory layer* supports the storage, retrieval, and continuous updating of historical knowledge to enable both short-term contextual awareness and long-term knowledge adaptation. The *reasoning and planning layer* analyzes perceived system states and maintains an internal world model to capture network dynamics and domain interactions. Based on this model, it translates high-level service intents into coordinated control plans through structured task decomposition and chain-of-thought (CoT) reasoning. Finally, the *action and tooling layer* executes the selected decisions through SAGIN control interfaces, orchestration platforms, and the direct actuation of ground, aerial, and space resources using external APIs and tools.

3) *Motivation for Cross-Domain Control in SAGIN*: Agentic AI provides a promising solution for enabling autonomous system operation with minimal human intervention by allowing intelligent entities to perceive, reason, plan, and act in dynamic environments. Meanwhile, 6G networks are evolving toward highly distributed, software-defined architectures characterized by heterogeneous control domains, such as the RAN and CN [12], which necessitate cross-domain orchestration. Specifically, cross-domain operation refers to coordinated decision-making and information exchange among these heterogeneous control domains, each of which may operate under different objectives, time scales, and resource constraints. Agentic AI-enabled cross-domain orchestration in 6G networks enables more adaptive and context-aware coordination across control domains. However, the large-scale and layered deployment in SAGIN introduces additional challenges, including highly dynamic topology, heterogeneous link characteristics, and multi-timescale control interactions, which significantly complicate cross-domain orchestration and make it difficult to satisfy the E2E service requirements of diverse SAGIN applications and services. In SAGIN environments,

aligning the objectives of different control domains to support SAGIN-assisted services becomes increasingly challenging due to heterogeneous computing capabilities and dynamically varying communication conditions across the ground, aerial, and space layers. Therefore, agentic AI-enabled cross-domain orchestration is required to coordinate and align the actions of agents operating across distributed control domains in SAGIN to meet E2E service requirements and system-level optimization goals under dynamic and heterogeneous network conditions.

III. AGENTIC AI-ENABLED CROSS-DOMAIN ORCHESTRATION FOR SAGIN

In this section, we introduce the Agentic AI-enabled cross-domain orchestration framework for SAGIN (AAI-CDOS).

A. Key Components AAI-CDOS

As shown on the right side of Fig. 1, AAI-CDOS consists of multiple cooperative agents deployed across different control domains, including the O-RAN domain, the computing infrastructure domain, and the E2E service domain. AAI-CDOS aims to optimize the Quality of Experience (QoE) and Quality of Service (QoS) of SAGIN-enabled applications such as maritime surveillance, disaster monitoring, and real-time video conferencing. These domains are logically associated with the ground, aerial, and space layers of the SAGIN architecture [1].

1) **E2E service Domain:** The E2E service domain is responsible for the E2E management of SAGIN-assisted services, such as intelligent transportation, disaster response, large-scale IoT monitoring, distributed learning, and immersive real-time services, across heterogeneous SAGIN layers. It interprets high-level service intents through a structured service intent descriptor, which formally captures application requirements in terms of service level agreement (SLA), QoS, QoE profile, and associated key performance indicators (KPIs) [12]. The service intent descriptor enables the translation of abstract service objectives into domain-specific policies and operational targets for RAN intelligent controllers (RICs) and computing orchestration controllers. Based on these intent specifications, the E2E service domain constructs service function graphs (SFGs) that define the ordered composition and placement of radio and computing functions across the ground, aerial, and space layers.

2) **CN domain:** The CN domain is responsible for mobility management, session control, policy enforcement, and user-plane traffic steering across heterogeneous SAGIN layers. The CN functions include the network slice subnet management function (NSSMF), policy control function (PCF), user plane function (UPF), network exposure function (NEF), session management function (SMF), network data repository function (NDRF), network slice selection function (NSSF), and access and mobility management function (AMF) [12]. These functions are typically deployed on ground infrastructure.

However, in SAGIN, the high orbital speeds of LEO satellites requires frequent network reselection and mobility management procedures, each involving multiple control signaling exchanges. In conventional ground-centric CN deployment,

these signals must traverse the ground CN and return to aerial or space nodes, resulting in increased round-trip delay. Therefore, CN functions can be partially distributed across aerial and space layers. Deploying lightweight or latency-sensitive CN components closer to satellites or UAVs reduces signaling delay, improves mobility responsiveness, and enhances service continuity in dynamic SAGIN environments.

3) **O-RAN domain:** RICs perform domain-local optimization of RAN functions, such as resource allocation, mobility management, and network slicing, by leveraging real-time network analytics and policy guidance from the CN. There are two RIC entities in the O-RAN architecture: the Near-Real-Time RIC (Near-RT RIC) and the Non-Real-Time RIC (Non-RT RIC). The Near-RT RIC controls RAN behavior at time scales ranging from 10 ms to 1 s and hosts xApps for functions such as radio scheduling, mobility management, and interference control. In contrast, the Non-RT RIC operates at time scales longer than 1 s and hosts rApps responsible for policy optimization, machine learning (ML) model training, and long-term RAN control [7]. In SAGIN, the O-RAN architecture can be conceptually divided into three logical layers with respect to RIC deployment:

- **Ground layer:** The ground layer hosts both the Near-RT RIC and the Non-RT RIC. The Near-RT RIC controls CU and DU functions through real-time RAN optimization, while the Non-RT RIC performs long-term policy optimization and analytics for ground radio access infrastructure and provides policy guidance to aerial nodes, such as BSs, RSUs, and UAVs.
- **Aerial layer:** The aerial layer consists of aerial computing and RAN platforms, such as UAVs, equipped with DUs and radio units (RUs). Due to the limited onboard computing and energy resources of battery-powered UAVs and HAPS, Near-RT and Non-RT RIC functions are deployed on ground infrastructure and provide remote control and policy guidance to aerial RAN components.
- **Space layer:** The space layer includes satellite-based RAN components operating under stringent constraints in onboard computation, communication, and power availability. Due to the high cost and limited availability of satellite resources, satellites typically support only lightweight or delegated RAN control functions, such as simplified Near-RT RIC functionalities co-located with the DU on GEO platforms and coordinated via inter-satellite links (ISLs) [7].

4) **Computing Infrastructure Domain:** The computing infrastructure domain coordinates heterogeneous computing nodes deployed across the ground, aerial, and space layers of SAGIN, including edge servers at base stations (BSs), UAVs, HAPS platforms, and satellites, as shown on the left side of Fig. 1. These nodes across different layers exhibit significant heterogeneity in terms of processing capability, energy availability, storage capacity, and communication latency, thereby requiring intelligent and adaptive resource management mechanisms. Moreover, due to the dynamic topology and resource variability of SAGIN, the computing infrastructure domain must operate under stringent constraints, including limited

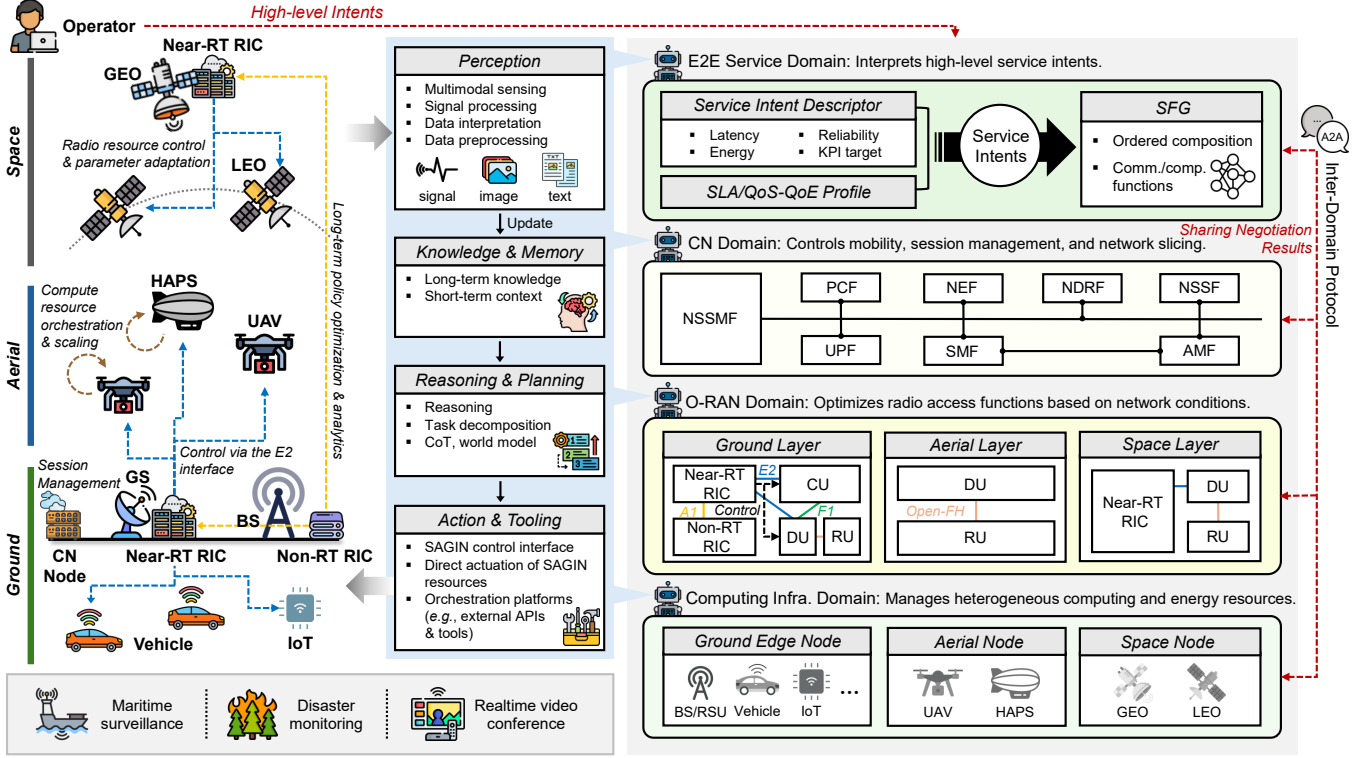


Fig. 1. Overview of AAI-CDOS. Through inter-domain negotiation and agent-to-agent coordination, the E2E service, CN, O-RAN, and computing infrastructure domains collaboratively optimize radio and computation resources across ground, aerial, and space layers. The agentic AI in each domain performs perception, knowledge management, reasoning, and action to dynamically align domain-level objectives with the E2E service requirements of various SAGIN applications.

onboard energy in UAVs and satellites, dynamic changes in link capacity, and connectivity instability resulting from mobility of the nodes. Therefore, the computing infrastructure domain continuously monitors resource states and adapts task placement decisions to maintain efficient resource utilization and service stability.

B. Interfaces and Interactions

AAI-CDOS leverages standardized O-RAN and 3GPP interfaces to enable coordination and information exchange among O-RAN and CN components, computing infrastructure, and E2E service management entities across SAGIN layers [13]. Within the O-RAN domain, the Near-RT RIC communicates with RAN nodes, including CUs and DUs, through the E2 interface, enabling Near-RT control and optimization of RAN functions. The F1 interface connects the CU and DU, while the Open FrontHaul (Open-FH) links the DU and RU, supporting flexible functional splits across ground, aerial, and space layers as illustrated in Fig. 1. The Non-RT RIC interacts with the Near-RT RIC through the A1 interface to deliver long-term policies, ML models, and optimization objectives.

C. Multi-Agent Communication in AAI-CDOS

Inter-domain communication facilitates collaboration and coordination among agents deployed across heterogeneous control domains through agent interaction mechanisms, including agent-to-agent (A2A) communication, the model context protocol (MCP), and IBM's agent communication protocol

(ACP) [14], [15]. Moreover, conventional, semantic communication mechanisms can be used to transmit task-relevant meanings and service intents rather than raw data, thereby improving coordination efficiency in terms of bandwidth utilization, signaling overhead, and decision latency across domains. While each domain independently optimizes its own objectives based on local information and constraints, the agents on different domain continuously negotiate and exchange information to ensure global consistency and alignment with E2E service requirements. In this process, inter-domain communication enables the monitoring and sharing of domain-level information, including traffic patterns, network load, and QoS metrics, thereby ensuring seamless operation of the SAGIN across multiple domains.

In AAI-CDOS, coordination and negotiation among agents across heterogeneous agents on cross-domain can be realized through several representative mechanisms [15]: In *Leader-Based* the E2E service domain agent can act as the leader to resolve conflicts among the CN, O-RAN, and computing infrastructure domains by arbitrating resource allocation priorities and aligning domain-level decisions with E2E service intents. Under *Consensus*, the agents operating in different domains coordinate through iterative message exchanges to reach a common agreement. In *Vertex Cover*, In AAI-CDOS, vertex cover mechanisms can be employed to select a small set of representative agents, such as regional RICs or orchestration agents, to oversee cross-domain interactions, reducing signaling overhead while preserving global visibility.

IV. CASE STUDY: AAI-CDOS FOR FEDERATED LEARNING

In this section, we present a case study demonstrating the effectiveness of the proposed AAI-CDOS in optimizing user experience for an E2E application over SAGIN. We consider an emergency model update scenario for wildfire detection using ground IoT sensors within a SAGIN environment. In the considered scenario, distributed ground sensor devices collect environmental data and periodically synchronize with a global model maintained at a satellite-based aggregation node, with the assistance of UAVs. Specifically, during winter conditions, the network provider requires urgent model updates to enhance wide-area wildfire detection accuracy and responsiveness. Ground IoT devices deployed in mountainous and forested areas continuously collect environmental data, while UAVs follow predefined trajectories to gather additional observations and perform local model updates based on the collected data. Upon completion of local training, the UAVs transmit their updated model parameters to LEO satellites acting as aggregation servers. Accordingly, these LEO satellites aggregate the local model updates received from the UAVs, while the UAVs collect data from ground IoT devices.

1) *Service Intent Initialization*: The FL process is initiated by the E2E service domain in response to the emergency wildfire detection task. High-level service intents are specified to prioritize rapid global model convergence under stringent latency constraints, high reliability requirements, and appropriate training periodicity. These service intents are translated into domain-specific policies and constraints and are disseminated to the O-RAN domain, the CN domain and the computing infrastructure domain deployed across the ground, aerial, and space layers through an inter-domain protocol.

2) *Context Observation and Analytics*: Upon receiving the service intents, RICs in the O-RAN domain and computing orchestration controllers in the computing infrastructure domain jointly perform continuous context observation across the SAGIN. The O-RAN domain monitors radio- and mobility-related states, including channel quality, link stability, spectrum utilization, traffic load, and the mobility patterns of UAVs and satellite-connected terminals. In parallel, the computing infrastructure domain observes computation-centric states such as local processing capability, energy availability, battery consumption rates, and task execution latency at ground IoT devices, UAVs, and satellite nodes.

3) *Cross-Domain Coordination and Negotiation*: During the client participation and model update transmission phase, cross-domain coordination and negotiation are performed among the O-RAN domain, CN domain, and the computing infrastructure domain. Due to heterogeneous resource constraints, conflicts may arise between communication-oriented and computation-oriented objectives during FL client selection. Specifically, the O-RAN domain tends to select UAVs with strong channel conditions and stable connectivity to both ground clients and satellite nodes, whereas the computing infrastructure domain prioritizes energy efficiency and battery preservation, particularly for UAVs operating under limited flight endurance.

To resolve these conflicts, the proposed AAI-CDOS framework enables inter-domain negotiation through coordinated decision-making among the O-RAN domain, CN domain and the computing infrastructure domain. For instance, when a UAV is identified as a strong communication relay but has limited remaining energy, the two domains negotiate a balanced participation strategy in which the UAV contributes partially to the FL process under adjusted operational constraints. Through such negotiation, communication performance and computational sustainability are jointly considered, allowing client selection and execution strategies to be adapted to satisfy overall system objectives.

4) *Agent Learning and Strategy Adaptation*: Under the AAI-CDOS framework, agents continuously adapt their decision strategies based on observed system outcomes during successive FL rounds. Agents update their policies for coordination, negotiation, and resource utilization using historical observations of link dynamics, UAV mobility patterns, and satellite processing performance. Through this iterative adaptation process, the system improves its ability to support rapid and reliable FL updates in highly dynamic SAGIN environments, enabling robust emergency response compared to independent domain optimization or static centralized control approaches.

A. Simulations

1) *Experimental Setup*: We conduct simulations on a Linux server running Ubuntu 22.04.2 LTS. The agent models operating in different domains, the E2E service domain, CN domain, O-RAN domain, and computing infrastructure domain, are each implemented using GPT-4o as illustrated in Fig. 1. The simulated environment models a UAV swarm with 10, 25, 50 UAVs, where UAVs collect sensor data from IoT devices deployed in mountainous areas and perform local training. The MNIST dataset is used as the benchmark dataset for evaluation. The operator input is shown in Fig. 2, while the specific deadline for FL is determined by AAI-CDOS based on network conditions—such as channel gain, received signal strength indicator (RSSI), propagation conditions including line-of-sight (LoS) and non-line-of-sight (NLoS), and signal-to-noise ratio (SNR), as well as the mission requirements of wildfire detection services, which are dynamically perceived and evaluated by the agents.

As baselines, we consider a *single-domain optimization scheme (Single Domain)*, where UAV participation is determined by a one control domain without considering the cross-domain constraints such as computational capacity or deadline feasibility [9], [10]. We also consider an *independent multi-domain scheme (Independent)*, in which each domain, including the RAN, computing infrastructure, CN, and E2E service domains, makes decisions independently based on its own locally defined objectives and information, without inter-domain coordination or joint objective alignment.

2) *Performance Analysis*: The left chart in the rightmost box of Fig. 2 shows the average number of selected UAVs and, out of a total of 10 UAVs, the number that successfully complete local training and upload their model parameters

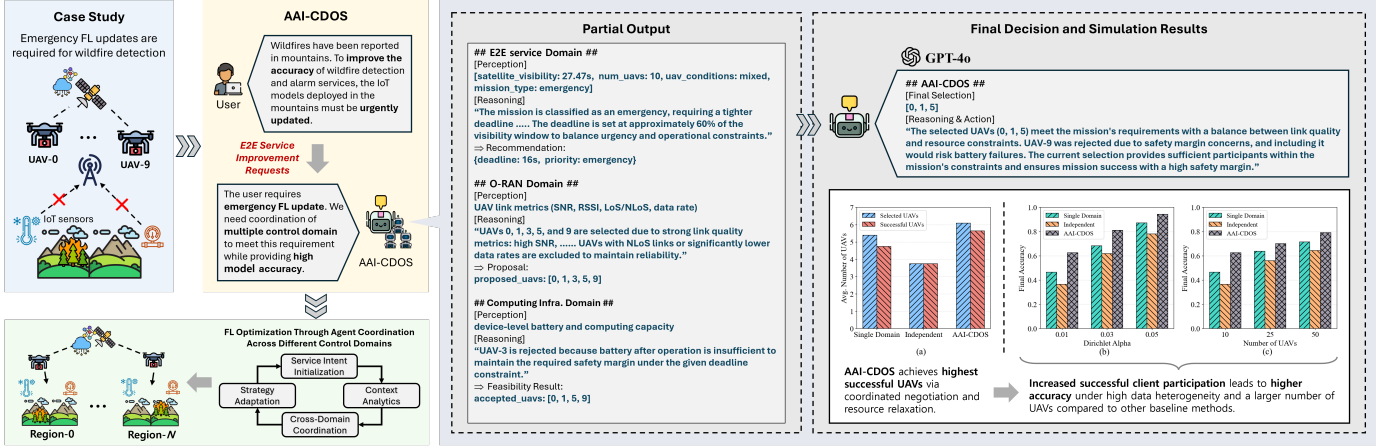


Fig. 2. Case study of AAI-CDOS for emergency FL in a wildfire detection scenario, where UAVs and IoT sensors are deployed across multiple regions and agent coordination is required across different control domains. AAI-CDOS processes the service improvement request through sequential agent reasoning and finalizes the UAV selection with justification. The evaluation results illustrate (a) the average number of UAVs selected for FL participation out of a total of 10 UAVs, and the number that successfully complete local training and parameter uploading within the given deadline, (b) the final MNIST test accuracy under different Dirichlet alpha settings, and (c) the final test accuracy under different numbers of total UAVs compared to benchmark schemes.

to the satellites within the deadline set by the E2E service domain. The proposed AAI-CDOS selects a larger number of UAVs while maintaining a high number of successful completions, thereby enabling more participating clients in FL compared to the other two benchmark schemes. The Single Domain initially selects a large number of UAVs based on radio information, which may optimize transmission performance. However, it fails to consider battery depletion time and limited computational capacity during local training, causing the selected UAVs to miss the deadline in the FL process. On the other hand, the Independent scheme results in the smallest pool of participating UAVs, as the E2E domain does not consider coordination among the RAN, CN, and computing infrastructure domains. This lack of cross-domain coordination limits information sharing, causing each domain to make overly conservative decisions.

The impact of these selection strategies is directly reflected in the FL accuracy. As shown in the center chart in the rightmost box of Fig. 2, which illustrates the test accuracy of the global model under different data distribution parameters (*i.e.*, Dirichlet alpha values), the proposed approach achieves the fastest convergence rate and the highest accuracy compared to the baseline schemes. With 10, 25, and 50 participating UAVs, AAI-CDOS achieves 34.2%, 9.9%, and 10.8% higher accuracy than Single Domain, respectively. Under the same UAV configurations, AAI-CDOS also achieves 70.8%, 25.1%, and 22.9% higher accuracy compared to Independent. These results demonstrate that AAI-CDOS can optimize E2E services through coordinated operation across control domains, thereby enhancing overall E2E service performance. Specifically, the improved global model achieved by AAI-CDOS can be distributed across wide geographic regions via satellites, enhancing the QoS of wildfire detection in mountainous areas.

V. OPEN CHALLENGES AND RESEARCH DIRECTIONS

A. Learning-Aware Cross-Domain Coordination Mechanisms

While coordination among agents across different domains enables goal alignment and cooperative decision-making, their

learning processes must also be carefully designed to ensure robust and adaptive cross-domain orchestration. In large-scale SAGIN environments, agents operating in heterogeneous domains may experience different time scales of hardware and software updates, communication delays, and data availability, leading to heterogeneous and potentially inconsistent learning dynamics. Future research should focus on learning-aware multi-agent coordination mechanisms that enable cross-domain model evolution, knowledge sharing, and policy adaptation in dynamic network environments.

B. Safety and Security in Cross-Domain Agent Interactions

While multi-agent communication protocols, such as A2A protocols, aim to ensure reliable interactions among agents, security and trust management remain critical challenges in large-scale agentic systems. In loosely governed agent ecosystems, risks such as impersonation, data exfiltration, task tampering, and unauthorized privilege escalation can easily arise, potentially undermining system reliability and safety. Another important direction for future work is the design of secure and trustworthy multi-agent coordination mechanisms that incorporate authentication, authorization, and auditability across heterogeneous SAGIN domains, while preserving the scalability and autonomy required for agentic AI-enabled network orchestration.

VI. CONCLUSION

In this paper, we presented an agentic AI-enabled network architecture for SAGIN, introducing the roles of key control domains, including the O-RAN, CN, computing infrastructure, and E2E service management, along with their associated challenges and limitations. We then proposed an agentic AI-enabled cross-domain orchestration framework for SAGIN, referred to as AAI-CDOS, which enables coordinated decision-making among agents operating across distributed control domains. Through case studies based on FL scenarios in SAGIN, we demonstrated that the proposed framework outperforms both benchmark methods, where agents in each domain

optimize their objectives independently without cross-domain coordination, and centralized optimization approaches.

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